PRACTICUM SYNOPSIS

Smart Shot: IoT Nerf Turret with Bluetooth Control

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योगः कर्मसु कौशलम्
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Table of Contents

S.No	Title	Page No.
1	Introduction	1
1.1	Introduction	1
2	Problem Statement and Feasibility	3
2.1	Problem Statement	3
2.2	Feasibility Study	4
2.2.1	Technical Feasibility	4
2.2.2	Operational Feasibility	5
2.2.3	Economic Feasibility	5
2.2.4	Potential Challenges & Solutions	5
2.2.5	Market Feasibility	6
2.2.6	Conclusion	6
3	Requirements	7
3.1	Hardware Components	7
3.2	Software Components	8
—	References	9

CHAPTER-1
INTRODUCTION

1.1 Introduction

The Bluetooth-enabled Nerf gun represents a creative and educational DIY project that combines mechanical engineering, electronics, and wireless control technology to produce a fun and interactive shooting system. This modified Nerf blaster introduces a remote-controlled mechanism that elevates the conventional toy gun into a smart, high-tech device [1].

At the heart of the system lies the Arduino Nano R3 microcontroller, a compact and versatile platform that manages all the core operations, including aiming and firing. The aiming mechanism utilizes a tilt-and-pan system, which is controlled by two MG90S micro servos. These metal gear servos are known for their precision and reliability, enabling smooth horizontal and vertical movement of the barrel. This setup allows users to accurately target their shots with minimal delay or jitter, making the experience much more engaging [2].

The dart-firing system is equally innovative. It consists of two Type 130 brushed DC motors that power a pair of spinning rollers. These rollers act as the primary propulsion system for the darts. To launch a dart, a third servo motor is used to push the dart from the magazine into the spinning rollers, which then shoot the dart forward with significant velocity. The magazine itself is capable of holding up to seven foam darts, providing semi-automatic functionality and enabling a rapid-fire sequence before a manual reload is needed [3].

Wireless control is implemented through an HC-05 Bluetooth module, which allows the user to operate the system remotely from a smartphone or computer. This wireless interface brings a new dimension of interactivity, letting users aim and fire the Nerf gun from a distance via a Bluetooth-compatible application [4]. Whether using a custom app or a terminal-based controller, the module provides a stable and responsive connection to the Arduino.

The electronic system incorporates several key components to ensure reliable and safe operation. Resistors of various values (330Ω, 680Ω, and 10kΩ) are employed to limit current, protect components, and ensure signal stability. For power management, an LM2596 buck converter is used to efficiently step down the voltage from the main power source to suitable levels for the motors and servos. In addition, an RFP30N06LE N-channel MOSFET handles the switching and control of the high-current motors, ensuring they operate only when necessary and preventing overheating or damage [5].

Other supporting elements include an FR207 rectifier diode, which protects against reverse voltage and current spikes, and Dupont connectors, which simplify wiring and make the system modular and easy to troubleshoot. Heat shrink tubing is used for electrical insulation, safeguarding exposed connections and reducing the risk of shorts. Structurally, a 1.5mm metal rod reinforces the assembly, contributing to the durability and stability of the design. A 20mm rubber band is also used in the construction, likely to maintain tension or aid in the magazine mechanism [6].

The entire system is powered by a 9V, 2.5A DC power supply, providing sufficient current to run the motors and servos without performance drops. The physical construction requires precise placement of components, secure fastening with screws, and organized wiring to maintain a compact form factor. The careful arrangement of these parts ensures both functionality and aesthetic appeal [7].

Ultimately, this Bluetooth-controlled Nerf turret is not only an entertaining project but also a valuable learning tool. It introduces key concepts in embedded systems, mechanical control, and wireless communication, making it ideal for hobbyists, students, and tinkerers. By blending traditional toy mechanics with modern technology, the project transforms an ordinary Nerf gun into a remote-operated, intelligent firing platform—highlighting the creative potential of electronics [8].

CHAPTER-2
PROBLEM STATEMENT AND FEASIBILITY

2.1 Problem Statement

Traditional guns lack technological features like remote control and automated mechanisms, limiting their functionality and appeal to enthusiasts seeking more dynamic, interactive experiences. This project addresses these limitations by designing a Bluetooth-enabled gun that incorporates remote tilt, pan, and firing controls, offering a modern and engaging alternative [1].

Additionally, current hobbyist-level turret systems rarely combine modular mechanical design, wireless control, and real-time actuation in a single, cohesive platform. This restricts their potential in educational applications where interdisciplinary skills such as programming, electronics, mechanical design, and user interface development must converge [2]. The absence of programmable features, feedback mechanisms, or intelligent control (e.g., automated targeting or integration with vision systems) further limits innovation and experimentation.

This project aims to overcome these shortcomings by designing and implementing a Bluetooth-enabled Nerf turret system that is remotely controllable via mobile or desktop applications. It will integrate pan and tilt mechanisms using servo motors, enable wireless dart firing, and support scalable features such as automated tracking in the future. The project not only serves recreational purposes but also acts as a pedagogical tool to teach core IoT concepts including microcontroller interfacing, sensor integration, wireless communication protocols, and power management. This practical implementation embodies the convergence of hardware and software, providing a robust solution to a common problem while fostering innovation in smart embedded systems [3].

2.2 Feasibility Study

A comprehensive feasibility study has been conducted to evaluate the practicality and viability of the proposed project. The assessment reveals that the system can be developed using widely available and cost-effective components such as the Arduino Nano microcontroller, HC-05 Bluetooth module, DC motors, and servo motors. These components are not only economical but also relatively easy to integrate and assemble, making the project particularly appealing to hobbyists, students, and educators [4]. The Bluetooth module enables seamless wireless communication with Android smartphones or personal computers, allowing for remote control of the system. Power is supplied via a 9V battery, though alternate options may be explored for higher efficiency. Despite the technical simplicity, the system presents a few challenges such as maintaining stable communication, achieving accurate mechanical alignment, and ensuring effective power distribution. These challenges, however, can be mitigated through careful testing, iterative improvements, and the use of appropriate supporting components [5].

2.2.1 Technical Feasibility

From a technical standpoint, the selected components are highly compatible and well-supported. The Arduino Nano microcontroller is capable of interfacing smoothly with the HC-05 Bluetooth module, standard servo motors, and DC motors. For efficient power delivery and control of motors, the use of a motor driver such as the L298N is recommended. This component not only helps regulate the power drawn by the motors but also safeguards the microcontroller from overload conditions. On the software side, development is facilitated by the Arduino Integrated Development Environment (IDE), which supports programming in C and C++. An extensive range of open-source libraries are available to assist with tasks such as serial communication (via the SoftwareSerial library), motor control (Servo, PWM, etc.), and Bluetooth connectivity. These factors collectively make the project technically sound and achievable even by those with a moderate level of electronics experience [6].

2.2.2 Operational Feasibility

Operational feasibility assesses the system's usability and ease of implementation. The assembly process is straightforward, relying on simple materials such as screws, plastic brackets, and rubber bands to mount the electronic and mechanical components. The user interface is designed to be intuitive, with control achieved through a Bluetooth connection to an Android application or PC software. This wireless feature adds convenience and flexibility, enabling the user to operate the system remotely. The ease of control and assembly significantly enhances the project's accessibility for educational demonstrations and DIY enthusiasts [7].

2.2.3 Economic Feasibility

Economically, the project is highly feasible due to its low component costs. The Arduino Nano, HC-05 Bluetooth module, and basic motors can all be sourced at a reasonable price, making the overall budget manageable. However, considerations must be made regarding the power source. While a standard 9V battery can power the microcontroller and low-power motors, it may not suffice for more demanding motor applications. In such cases, switching to a lithium-ion (Li-ion) battery pack or a regulated external power supply could provide better performance and longer operating time without significantly increasing the cost [8].

2.2.4 Potential Challenges & Solutions

Several challenges may arise during implementation, but each has practical solutions. One key concern is communication latency between the Bluetooth module and the microcontroller. This can be mitigated by optimizing baud rates and testing various configurations to minimize delay. Power management is another critical area; separating the power supplies for logic and motor circuits ensures that fluctuations in motor load do not interfere with microcontroller operation. Mechanical alignment of the moving parts, such as servos and DC motors, is essential for smooth and precise motion. This requires proper calibration and iterative testing of the mechanical system [9].

2.2.5 Market Feasibility

In terms of market applicability, the system is well-suited for learners, hobbyists, and educators who wish to explore robotics, electronics, or embedded systems. Its educational value is high due to the hands-on learning it promotes in both hardware and software development.

2.2.6 Conclusion

Overall, the feasibility study affirms that the proposed project is not only achievable but also cost-effective and educational. By leveraging readily available components, existing software libraries, and straightforward assembly methods, the project can be successfully implemented. With careful design, testing, and optimization, it holds significant potential for use in academic settings, DIY projects, and STEM learning environments.

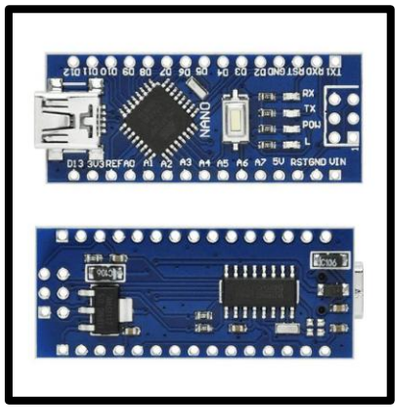
CHAPTER-3

REQUIREMENTS

3.1 Hardware Components

The hardware components selected for this project were chosen based on compatibility, cost-effectiveness, and ease of integration. Each component plays a critical role in ensuring the overall functionality and performance of the system. The table below provides a detailed overview of the key hardware elements used, including their specifications and purpose within the project [9].

S.no.	Parts	Quantity
1	1.5mm metal rod	1
2	2.1x5.5mm jack plug	1
3	wires	20
4	20mm rubber band	1
5	M2x10 self-taping screws	30
6	Arduino Nano R3	1
7	M5*20 countersunk screw	4
8	HC-05 Bluetooth module	1
9	MG-90s servos	3
10	LM2596 buck converter	1



Arduino Nano R3

LM2596 buck converter

3.2 Software Components

The software components form the backbone of the system's functionality, enabling communication, control, and coordination between the hardware elements. Development was carried out using the Arduino Integrated Development Environment (IDE), which supports C/C++ programming and provides a wide range of libraries for tasks such as Bluetooth communication, motor control, and sensor integration. Custom logic was implemented to handle data parsing, signal processing, and real-time control, ensuring seamless operation and responsiveness. The table below outlines the primary software tools and libraries utilized in the project.

S.no.	Name
1	Arduino IDE
2	Android App
3	Repository including all the python and UI files

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